

Factory Primer

Everything you need before starting Track 1 or Track 2

Three minutes. No manufacturing background required.

1. Quick response, in three paragraphs

Twenty years ago a clothing brand ordered 100,000 units nine months before the season and hoped. Zara proved you could instead put a small batch in stores, watch what sells for a week, and reorder the winners within a fortnight. Shein industrialised the idea, and most of the industry now runs some version of it.

For the factory this changes the shape of the work completely: instead of a dozen huge orders planned months ahead, there are **hundreds of small orders in the building at once**, and the most profitable ones — the reorders of whatever is selling — **arrive without warning and are due in one to two weeks**.

Two consequences drive both tracks. First, nobody can hold the whole picture in their head any more, so the general manager spends the day asking "*what is actually happening right now — and can we take this new order?*" — building their co-pilot is **Track 1**. Second, the factory does not own enough capacity for its peaks, so batches are constantly farmed out to outside workshops, and "*who should make this?*" has to be answered within the hour — that is **Track 2**.

2. The factory

SweaterCo makes knitwear for mall brands. Every order follows the same four steps:

KNITTING	→	ASSEMBLY	→	WASHING	→	PACKING
(machines make the panels)		(joining panels into a garment)		(wash & press)		(labels, boxes, final check)

That is all the manufacturing knowledge the project needs. The factory works Monday to Saturday, and every order carries a customer, a piece count, a due date, and its current stage.

3. The outside workshops (Track 2)

When the factory is full, batches are sent to one of **eight outside workshops** in the surrounding region. They differ in ways that matter: one is fast but sloppy, one is immaculate but tiny, one is cheap but slow. Each

has a profile card in `workshops.csv`, and before any batch can go anywhere, three questions must be answered: **can they make it** (not everyone is equipped for every product), **can they take it now** (capacity and current queue), and **are they allowed to** (one is suspended, one is on trial).

Today all three answers live in one dispatcher's head, and the choice is posted in a group chat with no record of the reasoning. Building the system that answers them instantly — and proving it allocates better than simple rules — is the whole of Track 2.

4. The data

Three small, clean CSV files (see `data_dictionary.md` — half a page):

File	What it is
<code>orders.csv</code>	120 orders: who, what, how many, due when, at what stage
<code>production_log.csv</code>	Daily output per stage for 90 days
<code>workshops.csv</code>	Profile cards for the 8 outside workshops: what they make, capacity, queue, quality, price, status

The data is intentionally simple — you can read all of it in a spreadsheet in five minutes. The hard part of these tracks is the agent and its interface, not the data.

5. Glossary

Term	Meaning
Quick response	Small first runs, then fast reorders of whatever sells
Reorder	A repeat of a style that is selling — urgent and unplanned
Order	One job in the factory
Stage	One of the four steps: knitting, assembly, washing, packing
Batch	An order (or part of one) sent to a workshop as one unit of work
Workshop / subcontractor	An outside shop that takes batches when the factory is full